

Exercise J - A

1. Consider the system of equations

$$x - 2y + 3z = -1, -x + y - 2z = k, x - 3y + 4z = 1$$

STATEMENT-1 : The system of equations has no solution for $k \neq 3$.
and

STATEMENT-2 : The determinant $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$, for $k \neq 3$.

[JEE-2008]

- (A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1
(B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

समीकरणों के निकाय

$$x - 2y + 3z = -1$$

$$-x + y - 2z = k$$

$$x - 3y + 4z = 1$$

कथन-1 : $k \neq 3$ के लिये समीकरणों के निकाय का कोई हल नहीं है।

तथा

कथन-2 : $k \neq 3$ के लिये सारणिक $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$ है।

- (A) कथन-1 सत्य है, कथन-2 सत्य है, कथन-2 कथन-1 का सही स्पष्टीकरण है
(B) कथन-1 सत्य है, कथन-2 सत्य है, कथन-2 कथन-1 का सही स्पष्टीकरण नहीं है
(C) कथन-1 सत्य है, कथन-2 असत्य है
(D) कथन-1 असत्य है, कथन-2 सत्य है

Ans. A

Sol. $x - 2y + 3z = -1, -x + y - 2z = k$
and $x - 3y + 4z = 1$

$$\Delta = \begin{vmatrix} 1 & -2 & 3 \\ -1 & 1 & -2 \\ 1 & -3 & 4 \end{vmatrix} = 0 \text{ and } \begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} = 3 - k$$

Hence if $k = 3$ then system will have infinite solutions and $k \neq 3$ then system will have no solution. so S(I) and S(II) both are true & (II) is correct explanation for (I).

2. The number of all possible values of θ , where $0 < \theta < \pi$, for which the system of equations

$$(y+z) \cos 3\theta = (xyz) \sin 3\theta$$

$$x \sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z}$$

$$(xyz) \sin 3\theta = (y+2z) \cos 3\theta + y \sin 3\theta$$

have a solution (x_0, y_0, z_0) with $y_0, z_0 \neq 0$, is.

[JEE-2010]

θ के उन सभी सम्भावित मानों की संख्या ज्ञात कीजिये, जिनके लिए समीकरण के निकाय

$$(y+z) \cos 3\theta = (xyz) \sin 3\theta$$

$$x \sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z}$$

$(xyz) \sin 3\theta = (y+2z) \cos 3\theta + y \sin 3\theta$ का एक हल (x_0, y_0, z_0) है, जहाँ $y_0, z_0 \neq 0$ तथा $0 < \theta < \pi$, है।

[JEE-2010]

Ans. 3

Sol. $(y+z) \cos 3\theta = (xyz) \sin 3\theta$... (i)

$$x \sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z}$$
 ... (ii)

$$(y+z) \cos 3\theta = (y+2z) \cos 3\theta + y \sin 3\theta$$
 ... (iii)

where $y, z \neq 0$ and $0 < \theta < \pi$

from (i) & (iii)

$$(y+z) \cos 3\theta = (y+2z) \cos 3\theta + y \sin 3\theta$$

$$\Rightarrow z \cos 3\theta + y \sin 3\theta = 0$$
 ... (iv)

from eqⁿ (ii)

$$2z \cos 3\theta + 2y \sin 3\theta = xyz \sin 3\theta$$
 ... (v)

from equation (iv) & (v)

$$\Rightarrow xyz \sin 3\theta = 0$$

$$\Rightarrow x \sin 3\theta = 0 \quad \text{as } yz \neq 0$$

Possible cases are either $x = 0$ or $\sin 3\theta = 0$

Case (1) : if $x = 0$

$$\Rightarrow y + z = 0 \Rightarrow y = -z$$

$$\text{from eqⁿ (iv) } \cos 3\theta = \sin 3\theta$$

$$\Rightarrow 3\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}$$

$$\Rightarrow \theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}$$

Case (2) : if $\sin 3\theta = 0$

$$\Rightarrow \theta = \frac{\pi}{3}, \frac{2\pi}{3}$$

But these values does not satisfy given equations.

Hence, total number of possible values of θ are 3.

3. Which of the following values of α satisfy the equation

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha?$$

[JEE-Adv.-2015]

α के निम्नलिखित मानों में कौन सा मान समीकरण

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha$$

को संतुष्ट करता है?

[JEE-Adv.-2015]

(A) -4

(B) 9

(C) -9

(D) 4

Ans. BC

Sol. $R_3 \rightarrow R_3 - R_2, R_2 \rightarrow R_2 - R_1$

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ 3+2\alpha & 3+4\alpha & 3+6\alpha \\ 5+2\alpha & 5+4\alpha & 5+6\alpha \end{vmatrix} = -648\alpha$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ 3+2\alpha & 3+4\alpha & 3+6\alpha \\ 2 & 2 & 2 \end{vmatrix} = -648\alpha$$

$$C_3 \rightarrow C_3 - C_2, C_2 \rightarrow C_2 - C_1$$

$$\begin{vmatrix} (1+\alpha)^2 & \alpha(2+3\alpha) & \alpha(2+5\alpha) \\ 3+2\alpha & 2\alpha & 2\alpha \\ 2 & 0 & 0 \end{vmatrix} = -648\alpha$$

$$\Rightarrow 2\alpha^2(2+3\alpha) - 2\alpha^2(2+5\alpha) = -324\alpha$$

$$\Rightarrow -4\alpha^3 = -324\alpha \Rightarrow \alpha = 0, \pm 9$$

4. The total number of distinct $x \in \mathbb{R}$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is. [JEE-Adv.-2016]

ऐसे सभी भिन्न $x \in \mathbb{R}$, जिनके लिए $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ है, की कुल संख्या है। [JEE-Adv.-2016]

Ans. 2

Sol. $\begin{vmatrix} x & x^2 & 1 \\ 2x & 4x^2 & 1 \\ 3x & 9x^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 2x & 4x^2 & 8x^3 \\ 3x & 9x^2 & 27x^3 \end{vmatrix} = 10$

Using $\begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} = (a-b)(b-c)(c-a)$

Again $\begin{vmatrix} x & x^2 & 1 \\ 2x & 4x^2 & 1 \\ 3x & 9x^2 & 1 \end{vmatrix} + (x)(2x)(3x) \begin{vmatrix} 1 & x & x^2 \\ 1 & 2x & 4x^2 \\ 1 & 3x & 9x^2 \end{vmatrix} = 10$

$\Rightarrow (x-2x)(2x-3x)(3x-x) + 6x^3[(x-2x)(2x-3x)(3x-x)] = 10$

$\Rightarrow 2x^3(1+6x^3) = 10$ (solve by putting $x^3 = t$)

$\Rightarrow x = -1, \left(\frac{5}{6}\right)^{\frac{1}{3}}$

5. Let $a, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations

$\alpha\xi + 2\psi = \lambda$

$3x - 2y = \mu$

Which of the following statement(s) is(are) correct ?

[JEE-Adv.-2016]

(A) If $\alpha = 3$, then the system has infinitely many solutions for all values of λ and μ

(B) If $\alpha \neq -3$, then the system has a unique solution for all values of λ and μ

(C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$

(D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$

माना कि $a, \lambda, \mu \in \mathbb{R}$ हैं। इन रैखिक समीकरणों के निकाय (system of linear equations) पर विचार कीजिए।

$$ax + 2y = \lambda$$

$$3x - 2y = \mu$$

निम्नलिखित में से कौन सा (से) कथन सही है (हैं)?

[JEE-Adv.-2016]

- (A) यदि $a = -3$, तब λ और μ के सभी मानों के लिए निकाय के अनन्त (infinitely many) हल हैं
 (B) यदि $a \neq -3$, तब λ और μ के सभी मानों के लिए निकाय का अद्वितीय (unique) हल है
 (C) यदि $\lambda + \mu = 0$, तब $a = -3$ के लिए निकाय के अनन्त हल हैं
 (D) यदि $\lambda + \mu \neq 0$, तब $a = -3$ के लिए निकाय का कोई हल नहीं है

Ans. BCD

Sol. System has unique solution for $\frac{a}{3} \neq \frac{2}{-2}$

system has infinitely many solution for $\frac{a}{3} = \frac{2}{-2} = \frac{\lambda}{\mu}$

and no solution for $\frac{a}{3} = \frac{2}{-2} \neq \frac{\lambda}{\mu}$

6. Let p, q, r be nonzero real numbers that are, respectively, the 10^{th} , 100^{th} and 1000^{th} terms of a harmonic progression. Consider the system of linear equations

[JEE(Adv.)2022]

$$x + y + z = 1$$

$$10x + 100y + 1000z = 0$$

$$qr x + pr y + pq z = 0$$

List - I

(I) If $\frac{q}{r} = 10$, then the system of linear equations has

(II) If $\frac{p}{r} \neq 100$, then the system of linear equations has

(III) If $\frac{p}{q} \neq 10$, then the system of linear equations has

(IV) If $\frac{p}{q} = 10$, then the system of linear equations has

List - II

(P) $x = 0, y = \frac{10}{9}, z = -\frac{1}{9}$

as a solution

(Q) $x = \frac{10}{9}, y = -\frac{1}{9}, z = 0$

as a solution

(R) infinitely many solutions

(S) no solution

(T) at least one solution

The correct option is:

माना कि p, q , एवं r शून्येतर वास्तविक संख्यायें हैं जो एक हरात्मक श्रेणी के क्रमशः 10वाँ, 100वाँ एवं 1000वाँ पद हैं। रैखिक समीकरणों के निकाय

[JEE Adv.-2022]

$$x + y + z = 1$$

$$10x + 100y + 1000z = 0$$

$$qr x + pr y + pq z = 0$$

पर विचार कीजिए।

सूची - I

(I) यदि $\frac{q}{r} = 10$ है, तब रैखिक समीकरणों के निकाय का

(II) यदि $\frac{p}{r} \neq 100$ है, तब रैखिक समीकरणों के निकाय का

(III) यदि $\frac{p}{q} \neq 10$ है, तब रैखिक समीकरणों के निकाय का

(IV) यदि $\frac{p}{q} = 10$ है, तब रैखिक समीकरणों के निकाय का

सूची - II

(P) हल $x = 0, y = \frac{10}{9}, z = -\frac{1}{9}$ है

(Q) हल $x = \frac{10}{9}, y = -\frac{1}{9}, z = 0$ है

(R) अनंत हल हैं

(S) कोई हल नहीं है

(T) कम से कम एक हल है

1. निम्नलिखित में से सही विकल्प चुनिए:

(A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (R); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T)

(B) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (R)

(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

(D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (T)

Ans. (B)

Sol. $x + y + z = 1$ _____ (1)

$$10x + 100y + 1000z = 0$$
 _____ (2)

$$qr x + pr y + pq z = 0$$
 _____ (3)

Equation (3) can be re-written as

$$\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 0 \quad (\because p, q, r \neq 0)$$

$$\text{Let } p = \frac{1}{a+9d}, q = \frac{1}{a+99d}, r = \frac{1}{a+999d}$$

Now, equation (3) is

$$(a+9d)x + (a+99d)y + (a+999d)z = 0$$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 10 & 100 & 1000 \\ a+9d & a+99d & a+999d \end{vmatrix} = 0$$

$$\Delta_x = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 100 & 1000 \\ 0 & a+99d & a+999d \end{vmatrix} = 900(d-a)$$

$$\Delta_y = \begin{vmatrix} 1 & 1 & 1 \\ 10 & 0 & 1000 \\ a+9d & 0 & a+999d \end{vmatrix} = 990(a-d)$$

$$\Delta_z = \begin{vmatrix} 1 & 1 & 1 \\ 10 & 100 & 0 \\ a+9d & a+99d & 0 \end{vmatrix} = 90(d-a)$$

Option I: If $\frac{q}{r} = 10 \Rightarrow a = d$

$$\Delta = \Delta_x = \Delta_y = \Delta_z = 0$$

And eq. (1) and eq. (2) represents non-parallel planes and eq. (2) and eq. (3) represents same plane
 $\Rightarrow$ Infinitely many solutions

I $\rightarrow$ P, Q, R, T

Option II: $\frac{p}{r} \neq 100 \Rightarrow a \neq d$

$$\Delta = 0, \Delta_x, \Delta_y, \Delta_z \neq 0$$

No solution

II $\rightarrow$ S

Option III: $\frac{p}{q} \neq 10 \Rightarrow a \neq d$

No solution

III $\rightarrow$ S

Option IV: If $\frac{p}{q} = 10 \Rightarrow a = d$

Infinitely many solution

IV $\rightarrow$ P, Q, R, T

7. Let α, β and γ be real numbers. Consider the following system of linear equations

$$x + 2y + z = 7$$

$$x + \alpha z = 11$$

$$2x + 3y + \beta z = \gamma$$

Match each entry in **List-I** to the correct entries in **List-II**.

List-I

List-II

(P) If $\beta = \frac{1}{2}(7\alpha - 3)$ and $\gamma = 28$, then the system has

(1) a unique solution

(Q) If $\beta = \frac{1}{2}(7\alpha - 3)$ and $\gamma \neq 28$, then the system has

(2) no solution

(R) If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where $\alpha = 1$ and $\gamma \neq 28$, then the system has

(3) infinitely many solutions

(S) If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where $\alpha = 1$ and $\gamma = 28$, then the system has

(4) $x = 11, y = -2$ and $z = 0$ as a solution

(5) $x = -15, y = 4$ and $z = 0$ as a solution

The correct option is :

[JEE(Adv.)2023]

माना कि α, β एवं γ वास्तविक संख्याएं (real numbers) हैं। निम्न रैखिक समीकरण निकाय (system of linear equation) पर विचार कीजिए।

$$x + 2y + z = 7$$

$$x + \alpha z = 11$$

$$2x + 3y + \beta z = \gamma$$

List-I की प्रत्येक प्रविष्टि (entry) का **List-II** की सही प्रविष्टियों (entries) से मिलान कीजिए।

List-I

List-II

(P) यदि $\beta = \frac{1}{2}(7\alpha - 3)$ एवं $\gamma = 28$, तब निकाय का(के)

(1) एक अद्वितीय हल (unique solution) है

(Q) यदि $\beta = \frac{1}{2}(7\alpha - 3)$ एवं $\gamma \neq 28$, तब निकाय का(के)

(2) कोई हल नहीं है

(R) यदि $\beta \neq \frac{1}{2}(7\alpha - 3)$ जहाँ $\alpha = 1$ एवं $\gamma \neq 28$, तब निकाय का(के)

(3) अनंत हल हैं

(S) यदि $\beta \neq \frac{1}{2}(7\alpha - 3)$ जहाँ $\alpha = 1$ एवं $\gamma = 28$, तब निकाय का(के)

(4) $x = 11, y = -2$ एवं $z = 0$ एक हल है

(5) $x = -15, y = 4$ एवं $z = 0$ एक हल है

1 ghfod Yf gS

[JEE(Adv.)2023]

(A) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (1); (S) $\rightarrow$ (4)

(B) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (5); (S) $\rightarrow$ (4)

(C) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (5)

(D) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (1); (S) $\rightarrow$ (3)

Ans. (A)

Sol. $D = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 0 & \alpha \\ 2 & -3 & \beta \end{vmatrix}$

$$= 1 [3\alpha] - 2 [\beta - 2\alpha] + [-3]$$

$$= 3\alpha - 2\beta + 4\alpha - 3$$

$$= 7\alpha - 2\beta - 3$$

$$\text{If } D = 0$$

$$7\alpha - 2\beta - 3 = 0$$

$$\beta = \frac{7\alpha - 3}{2}$$

$$D_x = \begin{vmatrix} 7 & 2 & 1 \\ 11 & 0 & \alpha \\ \gamma & -3 & \beta \end{vmatrix}$$

$$\text{if } \gamma = 28 \text{ \& } \beta = \frac{7\alpha - 3}{2}$$

$$\text{then } D_x = \begin{vmatrix} 7 & 2 & 1 \\ 11 & 0 & \alpha \\ 28 & -3 & \frac{7\alpha - 3}{2} \end{vmatrix}$$

$$7[3\alpha] - 2 \left[11 \frac{(7\alpha - 3)}{2} - 28\alpha \right] + [-33]$$

$$21\alpha - [11(7\alpha - 3) - 56\alpha] - 33$$

$$21\alpha - (77\alpha - 33 - 56\alpha) - 33$$

$$= 0$$

$$D_y = \begin{vmatrix} 1 & 7 & 1 \\ 1 & 11 & \alpha \\ 2 & \gamma & \beta \end{vmatrix}$$

$$\text{Again if } \gamma = 28 \text{ \& } \beta = \frac{7\alpha - 3}{2}$$

$$D_y = \begin{vmatrix} 1 & 7 & 1 \\ 1 & 11 & \alpha \\ 2 & 28 & \frac{7\alpha - 3}{2} \end{vmatrix}$$

$$R_1 \rightarrow R_1 - R_2$$

$$= \begin{vmatrix} 0 & -4 & 1-\alpha \\ 1 & 11 & \alpha \\ 2 & 28 & \frac{7\alpha-3}{2} \end{vmatrix}$$

$$4 \left[\frac{7\alpha-3}{2} - 2\alpha \right] + (1-\alpha)(28-22)$$

$$2(7\alpha-3) - 8\alpha + 6(1-\alpha)$$

$$14\alpha - 6 - 8\alpha + 6 - 6\alpha$$

$$= 0$$

$$D_z = \begin{vmatrix} 1 & 2 & 7 \\ 1 & 0 & 11 \\ 2 & -3 & \gamma \end{vmatrix}$$

$$1(33) - 2(\gamma - 22) + 7(-3)$$

$$33 + 44 - 2\gamma - 21 - 2\gamma$$

$$\Rightarrow 56 - 2\gamma$$

$$\text{If } \gamma = 28$$

$$D_z = 0$$

$$(P) \text{ If } \beta = \frac{1}{2}(7\alpha - 3) \text{ \& } \alpha = 28$$

then D_x, D_y, D_z are all zero

$\therefore$ Infinite many solution

$$P \rightarrow (3)$$

$$(Q) \text{ If } \beta = \frac{1}{2}(7\alpha - 3) \text{ \& } \gamma \neq 28$$

It can be easily observed that $D_z \neq 0$

$$\therefore D = 0 \text{ \& } D_z \neq 0$$

$\Rightarrow$ No solution

$$Q \rightarrow 2$$

$$(R) \text{ If } \beta \neq \frac{1}{2}(7\alpha - 3), \text{ i.e. } D \neq 0$$

It will be the case of unique solution but here additional info of $\alpha = 1$ \& $\gamma \neq 28$ is provided

$$(S) \text{ If } \alpha = 1 \text{ then}$$

$$x + 2y + z = 7$$

$$x + z = 11$$

$$2y = 7 - 11$$

$y = -2$ will always be the solution.

But $\gamma \neq 28$ then $z \neq 0$

$$\text{So, } R \rightarrow (1)$$

Again

$$\text{If } \alpha = 1; \gamma = 28$$

then $y = -2; z = 0$

& $z + 2(-2) = 7$

$x = 7 + 4$

$x = 11$

So we get the solution $S \rightarrow (4)$